TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMMS2613 PROBABILITY AND STATISTICS

TUESDAY, 7 OCTOBER 2025 TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

Question 1

a) The following data shows the number of customers entering a convenience store per hour during a business day.

62 45 57 69 50 39 64 53 48

(i) State the variable of interest for the above data and type of variable. (2 marks)

(ii) Calculate the median. (2 marks)

(iii) Calculate the mean and standard deviation. (5 marks)

b) A survey focusing on the number of hours employees spend on online training per month and the data is grouped as follows:

Time Spent (in nearest hour)	Number of employees
1 – 5	6
6 – 10	12
11 – 15	18
16 – 25	9
26 – 30	5

(i) Construct a "less than" ogive and estimate the (6 marks)

(1) median; (2 marks)

(2) percentage of employees spend more than 15 hours on the online training. (2 marks)

(ii) Calculate the mean and the standard deviation. (6 marks)

[Total: 25 marks]

Question 2

a) The table below shows the favorite modes of transportation among 240 college students. The options include Car, Bus, and Bicycle, and the data is grouped by gender.

Gender/Transportation Mode	Car (C)	Bus (B)	Bicycle (Y)	Total
Female (F)	45	50	25	120
Male (M)	55	30	35	120
Total	100	80	60	240

If a student is selected at random, find the probability that

(i) the student is male and prefers taking the bus, (2 marks)

(ii) the student prefers using the bus or is female. (3 marks)

(iii) the student preferred transport mode of bike given that the student is a female. (3 marks)

b) In a survey, 70% of employees say they prefer to attend online training sessions. Suppose 12 employees are selected at random. Find the probability that:

(i) fewer than two employees prefer offline training, (3 marks)

(ii) exactly 9 employees prefer online training. (3 marks)

c) The lifespans of a certain brand of smartphone batteries are normally distributed with a mean of 500 days and a standard deviation of 50 days.

(i) Find the probability that a randomly selected battery will last less than 580 days, (4 marks)

(ii) Find the probability that a randomly selected battery will last between 470 days and 520 days. (4 marks)

(iii) If 20% of the batteries are expected to last less than k days, find the value of k. (3 marks)

[Total: 25 marks]

Question 3

a) The weights of a particular breed of rabbits are normally distributed with a known standard deviation of 1.8 kg. A sample of 36 rabbits is taken, and the sample mean is found to be 4.5 kg.

Construct a 95% confidence interval for the true mean weight of the rabbits. (4 marks)

b) In a public opinion poll, 62 out of 120 randomly selected residents said they support the construction of a new community centre.

Construct a 90% confidence interval for the true proportion of residents who support the project. (4 marks)

c) A nutritionist claims that the average sugar content in a brand of energy drink is less than 10 grams. A random sample of 50 bottles gives an average sugar content of 9.3 grams with a standard deviation of 1.5 grams.

Test this claim at the 1% significance level. (8 marks)

d) Last year, 52% of high school students participated in co-curricular clubs. A recent survey of 90 students indicates that 60% of them are now participating in the activities being surveyed.

At the 5% significance level, test whether the proportion of students participating has changed. (9 marks)

[Total: 25 marks]

Question 4

a) The word "INFLATION" consists of 9 letters. Find the number of different arrangements that can be formed if:

(i) All letters are arranged in line. (2 marks)

(ii) The arrangements in part Question 4 a) (i) must begin with the letter 'I'. (2 marks)

(iii) The arrangements in part Question 4 a) (i) must begin with the letter 'A'. (2 marks)

b) The table below shows the marks obtained by 8 students in Linear Algebra and Java Programming tests:

Student	1	2	3	4	5	6	7	8
Marks in Linear Algebra (X)	45	80	70	55	90	60	50	85
Marks in Java Programming (Y)	48	75	68	52	85	60	55	78

(i) Calculate Pearson's product-moment correlation coefficient between marks in Linear Algebra and Java Programming. Interpret the result. (9 marks)

(ii) Find the least squares regression equation, $Y = a + bX$. (6 marks)

(iii) Estimate the marks in Java Programming for a student who scored 95 in Linear Algebra. Comment on the reliability of this prediction. (4 marks)

[Total: 25 marks]

AMMS2613 PROBABILITY AND STATISTICS

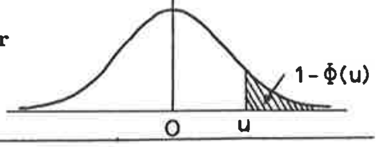
FORMULAE

Mode, grouped data $\text{Mode} = L_m + \dfrac{(f_m - f_b)}{2f_m - (f_b + f_a)} \times c_m$	**Median, grouped data** $\text{Median} = L_m + \dfrac{c_m}{f_m}\left(\dfrac{n}{2} - \sum f_{m-1}\right)$
Sample mean, raw data $\overline{X} = \dfrac{\sum x}{n}$	**Product moment coefficient of correlation** $r = \dfrac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$
Sample mean, grouped data $\overline{X} = \dfrac{\sum fx}{\sum f}$	**Spearman's rank correlation coefficient** $r_s = 1 - \dfrac{6\sum d^2}{n(n^2-1)}$
Population standard deviation, raw data $\sigma = \sqrt{\dfrac{\sum (x-\mu)^2}{N}}$ or $\sqrt{\dfrac{\sum x^2}{N} - \mu^2}$	**Least squares regression line:** $y = a + bx$ $b = \dfrac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$ $a = \dfrac{\sum y}{n} - b\dfrac{\sum x}{n}$
Sample standard deviation, raw data $s = \sqrt{\dfrac{\sum (x-\bar{x})^2}{n-1}}$ or $\sqrt{\dfrac{\sum x^2 - \dfrac{(\sum x)^2}{n}}{n-1}}$	
Population standard deviation, grouped data $\sigma = \sqrt{\dfrac{\sum fx^2}{\sum f} - \left(\dfrac{\sum fx}{\sum f}\right)^2}$	**Binomial Probability Function** $P(X = x) = {}^nC_x p^x (1-p)^{n-x}; \quad x = 0, 1, \cdots, n$
Sample standard deviation, grouped data $s = \sqrt{\dfrac{\sum fx^2 - \dfrac{(\sum fx)^2}{\sum f}}{\sum f - 1}}$	**Poisson Probability Function** $P(X = x) = \dfrac{e^{-\lambda}\lambda^x}{x!}; \quad \lambda > 0, \quad x = 0, 1, 2, \cdots$
Test Statistic for One Population $Z = \dfrac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}$ $Z = \dfrac{\overline{X} - \mu_0}{S/\sqrt{n}}$ (for big sample)	**Confidence Interval for One Population** $\overline{X} \pm Z_{\frac{\alpha}{2}} \dfrac{\sigma}{\sqrt{n}}$ $\overline{X} \pm Z_{\frac{\alpha}{2}} \dfrac{S}{\sqrt{n}}$ (for big sample)

$$Z = \frac{\hat{p}-\pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \qquad \hat{p} \pm Z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

AREAS IN TAIL OF THE NORMAL DISTRIBUTION

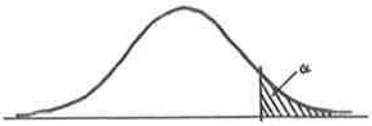
The function tabulated is $1 - \Phi(u)$ where $\Phi(u)$ is the cumulative distribution function of a standardised Normal variable u. Thus $1 - \Phi(u) = \frac{1}{\sqrt{2\pi}} \int_u^\infty e^{-u^2/2}\, du$ is the probability that a standardised Normal variable selected at random will be greater than a value of u $\left(= \frac{x-\mu}{\sigma}\right)$



$\frac{(x-\mu)}{\sigma}$	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641
0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
3.0	.00135									
3.1	.00097									
3.2	.00069									
3.3	.00048									
3.4	.00034									
3.5	.00023									
3.6	.00016									
3.7	.00011									
3,8	.00007									
3.9	.00005									
4.0	.00003									

PERCENTAGE POINTS OF THE NORMAL DISTRIBUTION

The table gives the 100α percentage points, u_α, of a standardised Normal distribution where $\alpha = \frac{1}{\sqrt{2\pi}} \int_{u_\alpha}^{\infty} e^{-u^2/2}\, du$. Thus u_α is the value of a standardised Normal variate which has probability α of being exceeded.



α	u_α	α	u_α	α	u_α	α	u_α	α	u_α	α	u_α
.50	0.0000	.050	1.6449	.030	1.8808	.020	2.0537	.010	2.3263	.050	1.6449
.45	0.1257	.048	1.6646	.029	1.8957	.019	2.0749	.009	2.3656	.010	2.3263
.40	0.2533	.046	1.6849	.028	1.9110	.018	2.0969	.008	2.4089	.001	3.0902
.35	0.3853	.044	1.7060	.027	1.9268	.017	2.1201	.007	2.4573	.0001	3.7190
.30	0.5244	.042	1.7279	.026	1.9431	.016	2.1444	.006	2.5121	.00001	4.2649
.25	0.6745	.040	1.7507	.025	1.9600	.015	2.1701	.005	2.5758	.025	1.9600
.20	0.8416	.038	1.7744	.024	1.9774	.014	2.1973	.004	2.6521	.005	2.5758
.15	1.0364	.036	1.7991	.023	1.9954	.013	2.2262	.003	2.7478	.0005	3.2905
.10	1.2816	.034	1.8250	.022	2.0141	.012	2.2571	.002	2.8782	.00005	3.8906
.05	1.6449	.032	1.8522	.021	2.0335	.011	2.2904	.001	3.0902	.000005	4.4172

Taken from Statistical Tables by J Murdoch and JA Barnes